Courier Management System - What System MUST Do

FR-01	**User Registration & Login** Customer, Delivery Agent, Admin can register/login with OTP verification
FR-02	**Parcel Booking** Sender enters receiver details, weight, size, pickup time slot & address
FR-03	**Cost Calculation** Auto-calculate shipping cost based on weight, distance, delivery type (Normal/Express)
FR-04	**Real-time Tracking** Generate unique Tracking ID (CMS123456), GPS tracking, status updates: Booked > Picked > In-Transit > Out for Delivery > Delivered
FR-05	**Proof of Delivery (POD)** Delivery agent captures photo / signature / OTP confirmation from receiver
FR-06	**Admin Dashboard** Admin can view all shipments, assign agents, generate invoices, view analytics

PHASE 2: REQUIREMENT ANALYSIS - NON-FUNCTIONAL & STAKEHOLDERS

Non-Functional Requirements

- **Performance:**
  Page load < 2 sec, Tracking API response < 1 sec
- **Security:**
  Password hashing, OTP expiry 5 min, JWT token
- **Usability:**
  Simple UI, 3-click booking, Mobile responsive
- **Scalability:**
  Handle 1000+ bookings/day, MySQL indexing
- **Availability:**
  99.5% uptime, Backup daily
- **Reliability:**
  SMS/Email alerts, No data loss for parcels

Stakeholders & Constraints

Customer (Sender/Receiver)

Book parcel, Track, Pay, Give feedback

Delivery Agent

Accept orders, Update status, POD

Admin / Manager

Manage all, Assign agent, Analytics, Billing

System Constraints

Budget low, 7 days timeline, Free SMS API (Twilio trial), No real GPS - Mock location

- Low budget, 7-day sprint
- Use MySQL not Firebase
- Mock GPS tracking
- OTP via Email (if SMS cost)

PHASE 2: REQUIREMENT ANALYSIS - USER STORIES & USE CASES

User Stories (Agile Format)

US-01	As a Customer, I want to book a courier in 3 clicks so that I can send parcel quickly
US-02	As a Customer, I want to track my parcel with ID so that I know where it is
US-03	As a Delivery Agent, I want to see assigned orders on my dashboard so that I can deliver
US-04	As a Delivery Agent, I want to update status to Delivered with OTP so that proof is stored
US-05	As an Admin, I want to view all bookings & generate invoice so that I manage business

Use Case Summary:

```
Customer -> [Register] [Login] [Book Parcel] [Track Parcel] [Make Payment] [View Invoice]
Delivery Agent -> [Login] [View Assigned Orders] [Update Status] [Upload POD Photo/Signature]
Admin -> [Login] [Manage Users] [Assign Agent] [View All Shipments] [Generate Reports] [Billing]
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